$$[T]_{SS'}\,u_4 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (1\ 0)^T$$

Q:- $T : \mathbb{R}^3 \to P_2$

$$T(a,b,c) = (a+b) + (b+c)t + (c+a)t^2$$

$$S = \{(1,1,1), (1,2,3), (0,-1,-1)\}$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{vmatrix} = 1(-2+3) - 1(-1+1) = 1 \neq 0$$

$$S' = \{1, 1+t, 1+t^2\}$$

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 1 \neq 0$$

Sol:

$$T(1,1,1) = 2 + 2t + 2t^2 = -2(1) + 2(1+t) + 2(1+t^2)$$

$$T(1,2,3) = 3 + 5t + 4t^2 = -6(1) + 5(1+t) + 4(1+t^2)$$

$$T(0,-1,-1) = -1 - 2t - t^2 = 2(1) - 2(1+t) - 1(1+t^2)$$

$$\alpha + \beta t + \gamma t^2 = a_0(1) + a_1(1+t) + a_2(1+t^2)$$

$$= (a_0 + a_1 + a_2) + a_1 t + a_2 t^2$$

$$a_0 + a_1 + a_2 = 2 \qquad a_0 = 3 - 5 - 4 = -6$$

$$a_1 = 2$$

$$a_2 = 2 \qquad a_0 = -1 + 2 + 1$$

$$\Rightarrow a_0 = -2$$

Pkalika Maths